

# PERCENTAGES

Short Revision Notes | Understand Level — Explain Ideas & Concepts

## 1. What is Percentage?

| Concept    | Explanation                     | Example                          |
|------------|---------------------------------|----------------------------------|
| Definition | "Percent" means "out of 100"    | 25% means 25 out of 100          |
| Symbol     | %                               | 50%, 75%, 100%                   |
| Origin     | From Latin "per centum"         | Per = for each, Centum = hundred |
| Meaning    | A fraction with denominator 100 | $25\% = 25/100 = 1/4$            |

### Understanding the Concept:

| Percentage | Meaning                     | Visual (out of whole) |
|------------|-----------------------------|-----------------------|
| 100%       | The whole                   | ■■■■■■■■■■            |
| 50%        | Half of the whole           | ■■■■■■■■■■            |
| 25%        | One quarter of the whole    | ■■■■■■■■■■            |
| 75%        | Three quarters of the whole | ■■■■■■■■■■            |
| 10%        | One tenth of the whole      | ■■■■■■■■■■            |

◆ **Key Understanding:** Percentage is a way of expressing a number as a fraction of 100. It makes comparisons easier.

## 2. Percentage, Fraction, Decimal

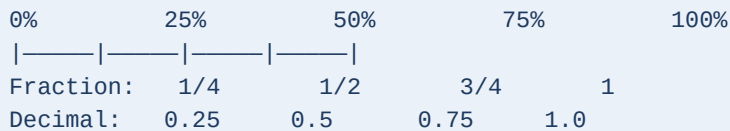
| Percentage | Fraction | Decimal | Meaning        |
|------------|----------|---------|----------------|
| 50%        | $1/2$    | 0.5     | Half           |
| 25%        | $1/4$    | 0.25    | Quarter        |
| 75%        | $3/4$    | 0.75    | Three quarters |
| 10%        | $1/10$   | 0.1     | One tenth      |
| 20%        | $1/5$    | 0.2     | One fifth      |
| 100%       | 1        | 1.0     | The whole      |

Percentage → Fraction: Divide by 100  $25\% = 25 \div 100 = 25/100 = 1/4$

Fraction → Percentage: Multiply by 100  $1/4 = 1/4 \times 100 = 25\%$

Percentage → Decimal: Divide by 100  $25\% = 25 \div 100 = 0.25$

Decimal → Percentage: Multiply by 100  $0.25 \times 100 = 25\%$



### 3. Understanding Profit

| Concept         | Explanation                                         | Example                                           |
|-----------------|-----------------------------------------------------|---------------------------------------------------|
| <b>Profit</b>   | When $SP > CP$ , the seller earns profit            | Buy for Rs 500, sell for Rs 650 → Profit = Rs 150 |
| <b>Why?</b>     | Seller adds a margin to cover costs and earn income | Production cost + Profit = Selling Price          |
| <b>Profit %</b> | Profit expressed as a percentage of Cost Price      | $(150 \div 500) \times 100 = 30\%$                |

| Cost Price    | Selling Price | Profit | Profit % |
|---------------|---------------|--------|----------|
| <b>Rs 100</b> | Rs 120        | Rs 20  | 20%      |
| <b>Rs 100</b> | Rs 150        | Rs 50  | 50%      |
| <b>Rs 200</b> | Rs 250        | Rs 50  | 25%      |

◆ **Key Understanding: Profit Percentage tells you how much profit you earn for every Rs 100 spent.**

If Profit% = 20%:

For every Rs 100 spent, you earn Rs 20 profit → Selling Price = Rs 120

$CP = 100 \rightarrow SP = 120$

$CP = 500 \rightarrow SP = 600$  (Profit = 100)

### 4. Understanding Loss

| Concept       | Explanation                                     | Example                                        |
|---------------|-------------------------------------------------|------------------------------------------------|
| <b>Loss</b>   | When $SP < CP$ , the seller incurs loss         | Buy for Rs 500, sell for Rs 450 → Loss = Rs 50 |
| <b>Why?</b>   | Items may be damaged, expired, or not in demand | Forced to sell at lower price                  |
| <b>Loss %</b> | Loss expressed as a percentage of Cost Price    | $(50 \div 500) \times 100 = 10\%$              |

| Cost Price    | Selling Price | Loss  | Loss % |
|---------------|---------------|-------|--------|
| <b>Rs 100</b> | Rs 80         | Rs 20 | 20%    |
| <b>Rs 100</b> | Rs 70         | Rs 30 | 30%    |
| <b>Rs 200</b> | Rs 180        | Rs 20 | 10%    |

◆ **Key Understanding: Loss Percentage tells you how much loss you incur for every Rs 100 spent.**

If Loss% = 10%:

For every Rs 100 spent, you lose Rs 10 → Selling Price = Rs 90

$CP = 100 \rightarrow SP = 90$

$CP = 500 \rightarrow SP = 450$  (Loss = 50)

## 5. Relationship: Profit % and CP/SP

### Relationship with Profit %

| Profit % | CP = 100 | SP = ? |
|----------|----------|--------|
| 0%       | 100      | 100    |
| 10%      | 100      | 110    |
| 20%      | 100      | 120    |
| 25%      | 100      | 125    |
| 50%      | 100      | 150    |
| 100%     | 100      | 200    |

For any CP:  $SP = CP \times (100 + \text{Profit}\%) \div 100$

If CP = 500 and Profit% = 10%

$SP = 500 \times 110 \div 100 = 550$

### Relationship with Loss %

| Loss % | CP = 100 | SP = ? |
|--------|----------|--------|
| 0%     | 100      | 100    |
| 10%    | 100      | 90     |
| 20%    | 100      | 80     |
| 25%    | 100      | 75     |
| 50%    | 100      | 50     |
| 100%   | 100      | 0      |

For any CP:  $SP = CP \times (100 - \text{Loss}\%) \div 100$

If CP = 500 and Loss% = 10%

$SP = 500 \times 90 \div 100 = 450$

## 6. Understanding Discount

| Concept                   | Explanation                          | Example                   |
|---------------------------|--------------------------------------|---------------------------|
| <b>Marked Price (MP)</b>  | The price printed on the item        | Rs 1000 on the tag        |
| <b>Discount</b>           | Reduction from marked price          | 20% off                   |
| <b>Why?</b>               | To attract customers, increase sales | "Sale!", "Festive Offer!" |
| <b>Selling Price (SP)</b> | Price after discount                 | MP – Discount             |

| Marked Price | Discount % | Discount Amount | Selling Price |
|--------------|------------|-----------------|---------------|
| Rs 1000      | 10%        | Rs 100          | Rs 900        |
| Rs 1000      | 20%        | Rs 200          | Rs 800        |

| Marked Price | Discount % | Discount Amount | Selling Price |
|--------------|------------|-----------------|---------------|
| Rs 1000      | 25%        | Rs 250          | Rs 750        |
| Rs 1000      | 50%        | Rs 500          | Rs 500        |

◆ **Key Understanding: Discount is ALWAYS calculated on the Marked Price, NOT on the Cost Price.**

$\text{Discount} = \text{MP} \times \text{Discount\%} \div 100$

$\text{SP} = \text{MP} - \text{Discount}$

$\text{SP} = \text{MP} \times (100 - \text{Discount\%}) \div 100$

## 7. Understanding Commission

| Concept           | Explanation                         | Example                     |
|-------------------|-------------------------------------|-----------------------------|
| <b>Commission</b> | Fee charged by a broker for service | 5% of transaction value     |
| <b>Why?</b>       | Payment for finding buyers/sellers  | Brokerage fee               |
| <b>Who Pays?</b>  | Usually seller pays commission      | Deducted from sale proceeds |

| Transaction Value | Commission % | Commission Amount | Amount Received |
|-------------------|--------------|-------------------|-----------------|
| Rs 1,000,000      | 5%           | Rs 50,000         | Rs 950,000      |
| Rs 500,000        | 3%           | Rs 15,000         | Rs 485,000      |
| Rs 200,000        | 10%          | Rs 20,000         | Rs 180,000      |

◆ **Key Understanding: Commission is ALWAYS calculated on the Transaction Value.**

$\text{Commission} = \text{Value} \times \text{Commission\%} \div 100$

$\text{Amount Received} = \text{Value} - \text{Commission}$

## 8. Relationship Between Profit, Discount & Marked Price

Cost Price (CP) → Marked Price (MP) → Selling Price (SP)

| Stage            | Calculation          | Example                                    |
|------------------|----------------------|--------------------------------------------|
| 1. Marked Price  | CP + Expected Profit | CP = 100, Expected Profit = 20% → MP = 120 |
| 2. Discount      | MP × Discount%       | Discount = 10% → 120 × 10% = 12            |
| 3. Selling Price | MP – Discount        | SP = 120 – 12 = 108                        |
| Final Profit     | SP – CP              | 108 – 100 = 8%                             |

### Complex Relationship Example:

A shopkeeper marks price 25% above CP, then gives a 10% discount.  
What is the actual profit?

$$CP = 100$$

$$MP = 100 + 25 = 125$$

$$\text{Discount} = 125 \times 10\% = 12.5$$

$$SP = 125 - 12.5 = 112.5$$

$$\text{Actual Profit} = 112.5 - 100 = 12.5\%$$

## 9. "Profit/Loss Percentage" Concept

| Concept  | Explanation                      | Example                                       |
|----------|----------------------------------|-----------------------------------------------|
| Profit % | Profit as a fraction of CP × 100 | $(\text{Profit} \div \text{CP}) \times 100\%$ |
| Loss %   | Loss as a fraction of CP × 100   | $(\text{Loss} \div \text{CP}) \times 100\%$   |
| Why CP?  | To show return on investment     | What you earn for each Rs 100 spent           |

| Profit/Loss % | Meaning                                   | If CP = Rs 100, SP = |
|---------------|-------------------------------------------|----------------------|
| 25% Profit    | For every Rs 100 spent, earn Rs 25 profit | Rs 125               |
| 25% Loss      | For every Rs 100 spent, lose Rs 25        | Rs 75                |
| 10% Profit    | For every Rs 100 spent, earn Rs 10 profit | Rs 110               |
| 10% Loss      | For every Rs 100 spent, lose Rs 10        | Rs 90                |

### ◆ Key Understanding: Profit/Loss Percentage makes it easy to compare different businesses.

Compare:

Business A: CP = 800, Profit = 100 →  $100/800 \times 100 = 12.5\%$

Business B: CP = 2500, Profit = 100 →  $100/2500 \times 100 = 4\%$

Business A is more profitable!

## 10. Understanding Discount Percentage

| Concept    | Explanation                                  | Example                                         |
|------------|----------------------------------------------|-------------------------------------------------|
| Discount % | Discount as a fraction of MP × 100           | $(\text{Discount} \div \text{MP}) \times 100\%$ |
| Why MP?    | To show how much reduction from marked price | "Save 20%!"                                     |

| Marked Price   | Discount | Discount % | You Pay |
|----------------|----------|------------|---------|
| <b>Rs 1000</b> | Rs 100   | 10%        | Rs 900  |
| <b>Rs 1000</b> | Rs 200   | 20%        | Rs 800  |
| <b>Rs 1000</b> | Rs 250   | 25%        | Rs 750  |

❖ **Key Understanding: Discount Percentage tells you how much you save for every Rs 100 of marked price.**

If Discount% = 20%:

For every Rs 100 of marked price, you save Rs 20 → You pay Rs 80

## 11. Understanding Commission Percentage

| Concept      | Explanation                                    | Example                                              |
|--------------|------------------------------------------------|------------------------------------------------------|
| Commission % | Commission as a fraction of Value $\times 100$ | $(\text{Commission} \div \text{Value}) \times 100\%$ |
| Why Value?   | Commission is based on sale value              | "5% of the sale price"                               |

| Sale Value   | Commission | Commission % |
|--------------|------------|--------------|
| Rs 1,000,000 | Rs 50,000  | 5%           |
| Rs 1,000,000 | Rs 30,000  | 3%           |
| Rs 500,000   | Rs 25,000  | 5%           |

◆ **Key Understanding: Commission Percentage** tells you how much broker charges for every Rs 100 of sale value.

If Commission% = 5%:

For every Rs 100 of sale value, broker charges Rs 5 → Seller receives Rs 95

## 12. Difference Between Marked Price, Cost Price & Selling Price

| Price Type         | Meaning                           | Formula                                               |
|--------------------|-----------------------------------|-------------------------------------------------------|
| Cost Price (CP)    | What the seller paid for the item | Given / Calculated                                    |
| Marked Price (MP)  | Price printed/written on item     | $CP + \text{Profit expectation}$                      |
| Selling Price (SP) | Actual price the customer pays    | $MP - \text{Discount, or } CP \pm \text{Profit/Loss}$ |

Step 1: Seller buys item → Cost Price (CP)

Step 2: Seller marks up price → Marked Price (MP)

Step 3: Seller offers discount → Selling Price (SP)

Step 4: Compare SP with CP → Profit or Loss

**Complete Example:**

| Stage           | Value   | Calculation            |
|-----------------|---------|------------------------|
| CP              | Rs 800  | Given                  |
| MP              | Rs 1000 | 25% above CP           |
| Discount %      | 10%     | Given                  |
| Discount Amount | Rs 100  | 10% of 1000            |
| SP              | Rs 900  | $1000 - 100$           |
| Profit          | Rs 100  | $900 - 800$            |
| Profit %        | 12.5%   | $100/800 \times 100\%$ |

### 13. Common Misunderstandings to Avoid

| Misunderstanding                     | Why It's Wrong                           | Correct Understanding                                                      |
|--------------------------------------|------------------------------------------|----------------------------------------------------------------------------|
| <b>Profit% = profit ÷ SP</b>         | Profit% is based on CP, not SP           | $\text{Profit\%} = (\text{Profit} \div \text{CP}) \times 100\%$            |
| <b>Loss% = loss ÷ SP</b>             | Loss% is based on CP, not SP             | $\text{Loss\%} = (\text{Loss} \div \text{CP}) \times 100\%$                |
| <b>Discount% = discount ÷ CP</b>     | Discount is based on MP, not CP          | $\text{Discount\%} = (\text{Discount} \div \text{MP}) \times 100\%$        |
| <b>Commission% = commission ÷ CP</b> | Commission is based on transaction value | $\text{Commission\%} = (\text{Commission} \div \text{Value}) \times 100\%$ |
| <b>Marked Price = Cost Price</b>     | MP includes expected profit              | $\text{MP} = \text{CP} + \text{Expected Profit}$                           |
| <b>Selling Price = Marked Price</b>  | SP may have discount applied             | $\text{SP} = \text{MP} - \text{Discount}$                                  |

### 14. Relationship Between Profit % and Discount %

| Scenario                                | Calculation    | Result         |
|-----------------------------------------|----------------|----------------|
| <b>CP=100, Markup=20%, Discount=0%</b>  | MP=120, SP=120 | Profit = 20%   |
| <b>CP=100, Markup=20%, Discount=20%</b> | MP=120, SP=96  | Loss = 4%      |
| <b>CP=100, Markup=25%, Discount=20%</b> | MP=125, SP=100 | No profit/loss |
| <b>CP=100, Markup=50%, Discount=20%</b> | MP=150, SP=120 | Profit = 20%   |

If Markup% = Discount% → Loss (not break-even!)  
 Because: CP 100 → MP 120 → SP after 20% discount = 96  
 20% markup and 20% discount = 4% loss!

### 15. Relationship Between Commission and Net Amount

| Concept            | Example        | Understanding        |
|--------------------|----------------|----------------------|
| <b>Gross Value</b> | Rs 1,000,000   | Total sale value     |
| <b>Commission</b>  | 5% = Rs 50,000 | Broker's fee         |
| <b>Net Amount</b>  | Rs 950,000     | What seller receives |

$\text{Commission} = \text{Value} \times \text{Commission\%} \div 100$   
 $\text{Net Amount} = \text{Value} - \text{Commission}$   
 $\text{Net Amount} = \text{Value} \times (100 - \text{Commission\%}) \div 100$

| Commission % | For Rs 100 Value | Net Amount |
|--------------|------------------|------------|
| <b>0%</b>    | Rs 0             | Rs 100     |
| <b>5%</b>    | Rs 5             | Rs 95      |
| <b>10%</b>   | Rs 10            | Rs 90      |
| <b>20%</b>   | Rs 20            | Rs 80      |



## 16. Check Your Understanding

| No. | Question                                       | Explain                                                        |
|-----|------------------------------------------------|----------------------------------------------------------------|
| 1   | What does "percent" mean?                      | Out of 100                                                     |
| 2   | Why is profit% based on CP?                    | To show return on investment                                   |
| 3   | Why is discount% based on MP?                  | To show reduction from marked price                            |
| 4   | Why is commission% based on value?             | To show fee for the service                                    |
| 5   | Can 20% markup and 20% discount give profit?   | No, it gives a loss (4% loss)                                  |
| 6   | What is the difference between CP and MP?      | MP includes expected profit                                    |
| 7   | What is the difference between MP and SP?      | SP may have discount applied                                   |
| 8   | Why is CP important in profit/loss?            | It's the base for calculating percentages                      |
| 9   | How do you compare two businesses with profit? | Use profit percentage ( $\text{profit}/\text{CP} \times 100$ ) |
| 10  | Why do shops offer discounts?                  | To attract customers and increase sales                        |

## 17. Key Understandings to Remember

- ★ PERCENTAGE IS "OUT OF 100".
- ★ PROFIT% AND LOSS% ARE BASED ON COST PRICE.
- ★ DISCOUNT% IS BASED ON MARKED PRICE.
- ★ COMMISSION% IS BASED ON TRANSACTION VALUE.
- ★  $\text{PROFIT} = \text{SP} - \text{CP}$  (when  $\text{SP} > \text{CP}$ )
- ★  $\text{LOSS} = \text{CP} - \text{SP}$  (when  $\text{SP} < \text{CP}$ )
- ★  $\text{DISCOUNT} = \text{MP} - \text{SP}$
- ★  $\text{MARKED PRICE} = \text{CP} + \text{EXPECTED PROFIT}$
- ★ 20% MARKUP + 20% DISCOUNT = 4% LOSS!
- ★ COMPARE BUSINESSES USING PERCENTAGES, NOT ABSOLUTE VALUES.

*Revision Tip: Understand the relationships between CP, MP, SP, Profit, Loss, and Discount. These are the building blocks of all percentage problems!*